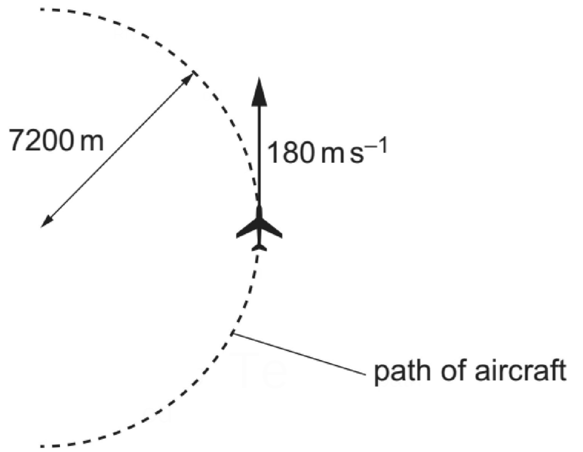
1 **(a)** With reference to velocity and acceleration, describe uniform circular motion.

....

....

.... [2]

(b) An aircraft is travelling in a horizontal circle of radius 7200 m at a speed of 180 m s^{-1}, as shown in the view from above in Figure 1.1.



not to scale

Figure 1.1

In order to move in a circle, the aircraft has to tilt its wings so that the lift force from the wings is not vertical, as shown in the view from behind in Figure 1.2.

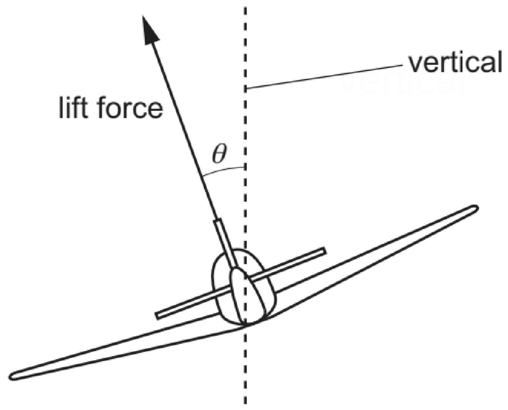


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Figure 1.2

(i) With reference to the forces acting on the aircraft, explain why the circular motion of the aircraft requires the lift force to be in the direction shown rather than vertically upwards.

....

....

.... [2]

(ii) On Figure 1.2, draw an arrow from the aircraft to represent the direction of the resultant force on the aircraft. [1]

(iii) Calculate the centripetal acceleration of the aircraft.

acceleration = ... $m\,s^{-2}$ [2]

(iv) Determine the angle θ at which the aircraft is tilted. Explain your reasoning.

θ = ... ° [3]

[Total: 10]